Introduction to Probability, Examples

CSU, Chico Math 314

2025-09-08

outline

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Set Operations

Conditional Probability

Mutual Independence

System Reliability

Association Rules

Set Operations Functions in R

The functions for set operations in R are as follows:

```
union(x,y) # combined elements
intersect(x,y) # shared elements
setdiff(x,y) # elements in x but not y (order matters)
setequal(x,y) # are the sets equal?
```

Note that these require you to have specific sets of objects (such as `x <- c(1:3)`). If you have probabilities instead, you will not want to use these functions.

Set Operations, homework (Q1)

Suppose you have a vector of dates of November 2024 (let's call it `march`), and that you have two other vectors of indices `weekends` and `break`, which store the dates of November's weekends and autumn break, respectively. Use R's built-in set operations on these vectors to print the dates of **non-vacation weekdays**.

```
november <- 1:30 # November 2025 dates
weekends <- c(1,2,8,9,15,16,22,23,29,30)# weekends
autumnbreak <- 24:28
## Hint: ?union
```

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Conditional Probability, example

Four cards are to be dealt successively at random and without replacement from a standard deck of cards. Find the probability of drawing the following: a club, then a heart, then a spade, and then a diamond.

Conditional Probability, homework (Q2)

In a game of chance, there are 18 cards on a table, 4 of which are yellow, 8 red, and 6 blue. The cards are placed facedown, and drawing a pair of matching cards wins a prize. Assume you draw two facedown cards. Let $P(1_{yellow})$ be the probability that the first card is yellow, and $P(2_{yellow})$ be the probability that the second card drawn is yellow.

Given that your first card drawn is yellow, what is the probability that the next card drawn will also be yellow, winning a prize? Hence, find $P(2_{yellow}|1_{yellow})$.

Additionally, before any cards have been drawn, what is the probability of winning a prize due to both cards being yellow? $(P(1_{yellow} \cap 2_{yellow}))$

Conditional Probability, homework (Q3)

From the *Lec02_Intro_Stats.pdf* notes, we saw probabilities for various combinations of traffic lights on slide 39.

- $P(A) = 0.55.$
- $P(A^c) = 0.45$
- $P(B|A) = 0.75$
- $P(B|A^c) = 0.09$
- $P(B) = 0.453$

where $A =$ first light is green, and $B =$ second light is green. Calculate $P(A^c|B)$ (the probability that the first light is NOT green given that the second light is green) and show your work.

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Mutual Independence, example

Roll a six-sided die once. Let $E_1 = \{1, 2, 3\}, E_2 = \{3, 4, 5\}$, and $E_3 = \{1, 2, 3, 4\}$. Does

$$P(E_1 \cap E_2 \cap E_3) = P(E_1)P(E_2)P(E_3)?$$

Are E_1 and E_2 independent?

Mutual Independence, homework (Q4A)

Roll a ten-sided die once. Let $E_1 = \{5, 6\}$ and $E_2 = \{1, 3, 5, 7, 9\}$. Are E_1 and E_2 independent? Show your calculations.

Mutual Independence, homework (Q4B)

Roll a ten-sided die once. Let $E_1 = \{3, 4, 5, 6\}$ and $E_2 = \{5, 6, 7, 8, 9, 10\}$. Are E_1 and E_2 independent? Show your calculations.

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System Reliability, motivation

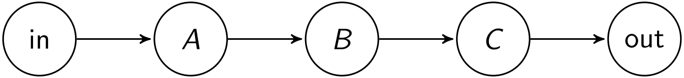

Figure 1: Sequential system

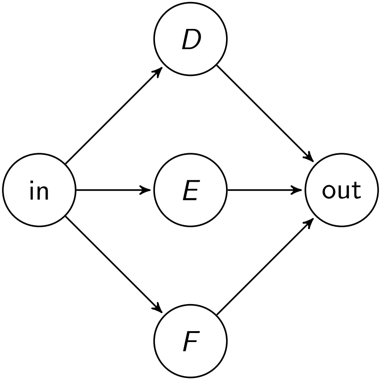

Figure 2: Parallel system

System Reliability, example 1

The three components of the sequential system, A, B, and C, will fail with probabilities $p_A = 0.04, p_B = 0.08$, and $p_C = 0.12$, independently of each other. What is the probability the sequential system will fail?

System Reliability, example 1: Reframing the question

If one part of the system fails, the whole system fails.

Rather than finding every combination of failure states, why not focus on the one state that DOES work: $P(\text{no component fails})$

Thanks to set operations, this can be a much quicker approach.

System Reliability, homework (Q5)

The three components of the parallel system (slide 14), D, E, and F, will function with probabilities $p_D = 0.80, p_E = 0.90$, and $p_F = 0.85$, independently of each other. What is the probability the parallel system will function?

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Association Rules, homework (Q6)

Consider the dataset `arules::Groceries`. Also import the CSV version from the listed URL (after removing the space).

```
suppressMessages(library(arules))
data(Groceries)
arGroceries <- apriori(Groceries,
           parameter=list(supp=0.001, conf=0.8))
inspect(head(arGroceries, n=6))
# Import from CSV version from URL
url <- "https://raw.githubusercontent.com/njlytal/
MATH314/main/groceries.csv"
groceries <- read.csv(url)
```

Calculate the support, confidence, and lift for row 3 and row 4 using the rules developed in the Lecture 02 notes. Use mean(), &, and | to do so, and use complete sentences to interpret the numbers you get. Check your work with `arGroceries`.

Homework (Q6) Tips

Note that the left hand side (LHS) of each rule is the **intersection** of the two items separated by commas. Your set operation rules may be useful here for setting up your statements in RStudio.

Also, note that the information within arGroceries is only used to check your work. The groceries.csv file contains the actual data you will be analyzing, where each row is a different customer and each column is a different grocery item.

references I

Michael G. Akritas. *Probability and Statistics with R for Engineers and Scientists.* Pearson Education, Inc., 2016.

Adapted from notes by Dr. Roualdes.